Due March 31 at the start of class 1:50 pm

Check ecampus before class on Tuesday March 31 for the next set of lecture and HW handouts. You must print these out and bring them to class.

Submit the following.

- Complete pages HW – 47 thru 51.

- Complete the MML assignments that are due.

MATH 2413 - Derivatives of Quotients Due Date:__________ HW__47

Classwork / Homework

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1 - 5. Use the quotient rule to find the first order derivative of each of the functions. Show all work and simplify completely.

1. Let $f(a) = \dfrac{3a^3 - 1}{2 - a^3}$.

$H(a) =$ $H(a) =$

$H'(a) =$ $L'(a) =$

2. Let $k(y) = \dfrac{2y^3}{3 - y^2}$.

Give the answer in factored form.

$H(y) =$

$L(y) =$

3. Let $k(x) = \dfrac{e^{2x}}{6 - x^2}$.

Give the answer in factored form.

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(continued) 1 - 5. Use the quotient rule to find the first order derivative of each of the functions. Show all work and simplify.

4. Let $f(x) = \dfrac{x^2}{x - \cos x}$.
Give the answer in factored form.

5. Let $f(\theta) = \dfrac{\cot\theta + 1}{\sin\ \theta}$.

Use trigonometric identities and LCD to simplify the numerator. Show all work in detail.

MATH 2413 - Higher Order Derivatives Due Date:__________ HW__49
Classwork / Homework

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1 - 5. Find the second order derivatives for each of the following. Show work using rigorous differential form. Give answers using the shorthand prime notation.

1. $f(x) = 2x^3 + 5x^2 - \frac{2}{x^3}$

2. $f(x) = (8x - 3)^{1/2}$

3. $f(x) = e^{x^3}$

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4. $f(x) = \sin(x^2) - e^{-3x}$

5. $f(x) = \csc(3x) - \cot(3x)$

MATH 2413 - Higher Order Derivatives Due Date:__________ HW__51
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Third and higher order derivatives use the following notations:

Third-order derivative of $f(x)$: $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{df}{dx}\right)\right) = \frac{d}{dx}\left(\frac{d^2 f}{dx^2}\right) = \frac{d^3 f}{dx^3} = f'''(x) = f^{(3)}(x)$

Fourth-order derivative of $f(x)$: $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{df}{dx}\right)\right)\right) = \frac{d^4 f}{dx^4} = f^{(4)}(x)$ etc.

5. Find the fifth-order derivative for $f(x) = \frac{x^6}{12} + 3x^2 - \cos x$. Show work in the rigorous differential notation. Give the answer using the numerical shorthand notation.

6. Find the fourth-order derivative for $g(x) = (2x - 12)^5$. Give answers and show work in rigorous differential form.